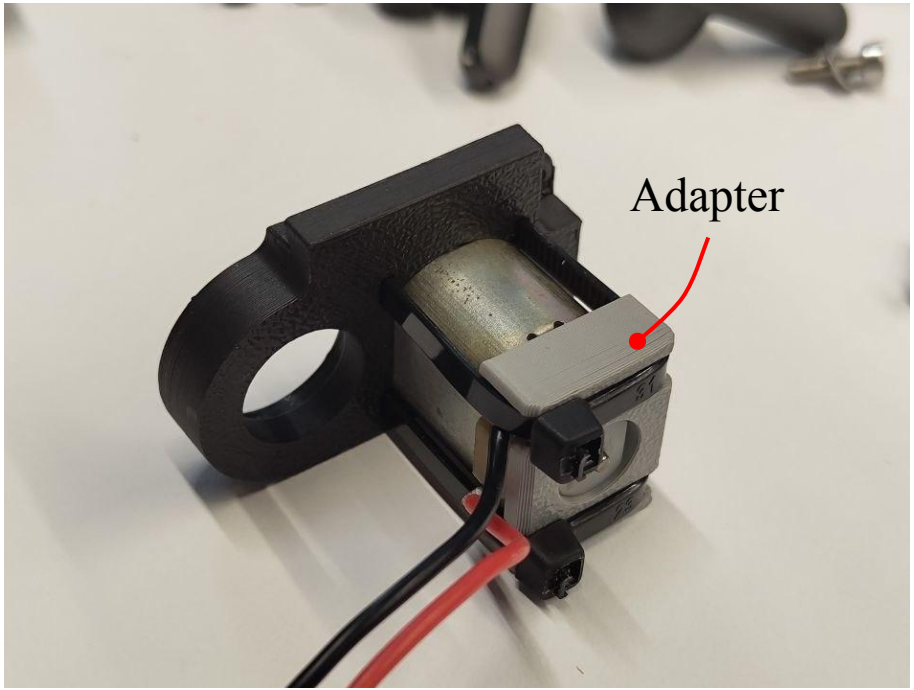
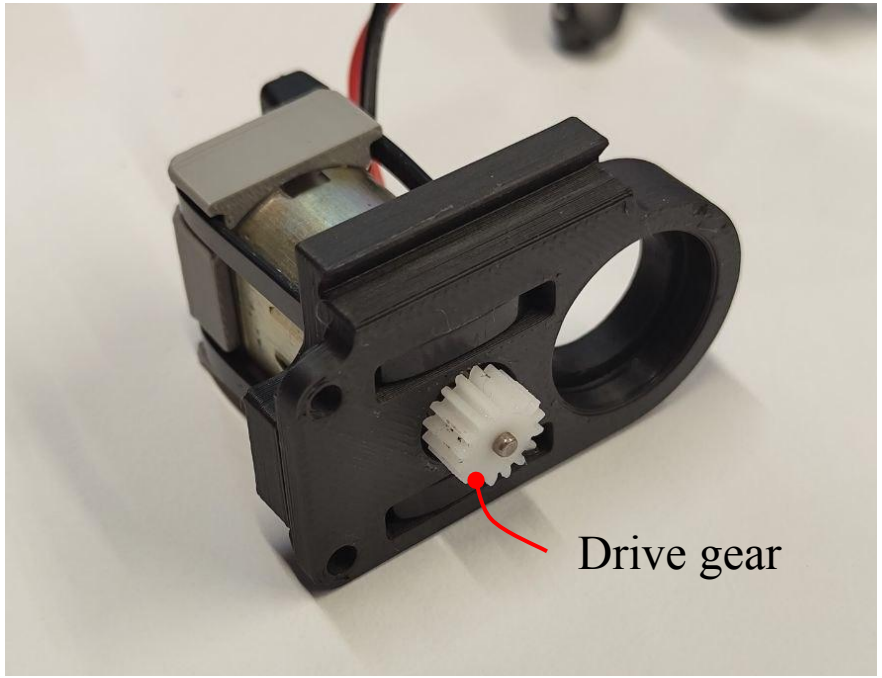
# Assembly guide

1.



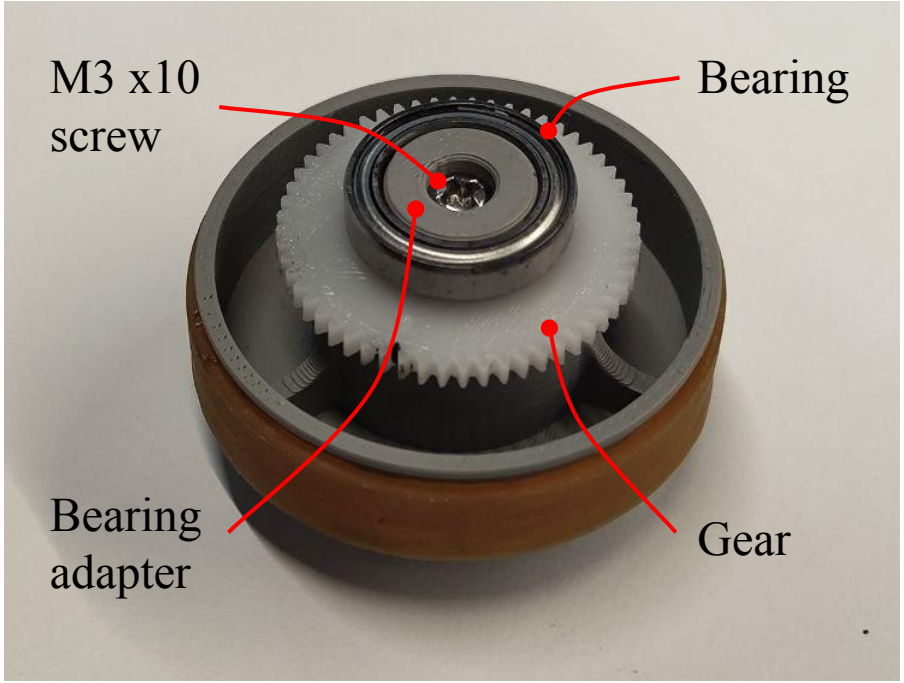

Insert DC motor into housing and place adapter at motor rear before using zip-tie to secure.

2.



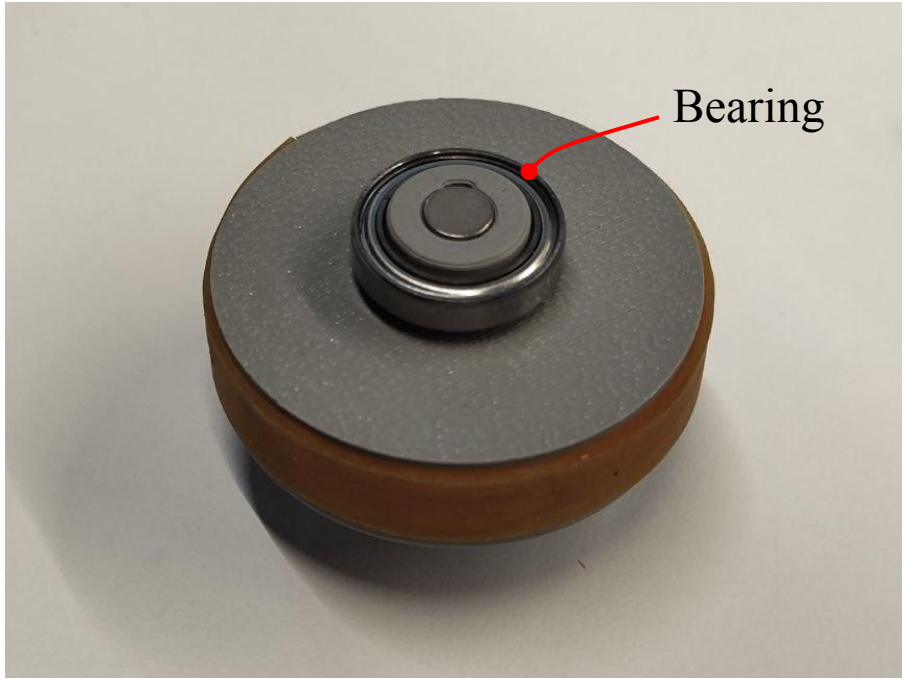

Ensure zip-tie passes through grooves on opposite side, and press-fit drive gear on motor shaft.

3.



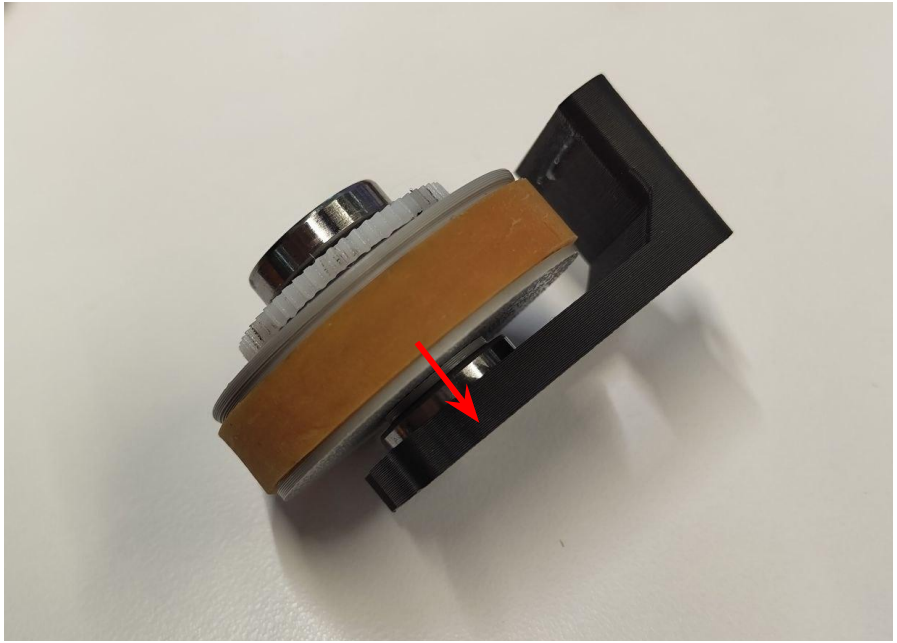

Attach gear & bearing adapter piece to flywheel and screw down with M3 x10mm screw. Press fit 17x30x7mm bearing.

4.



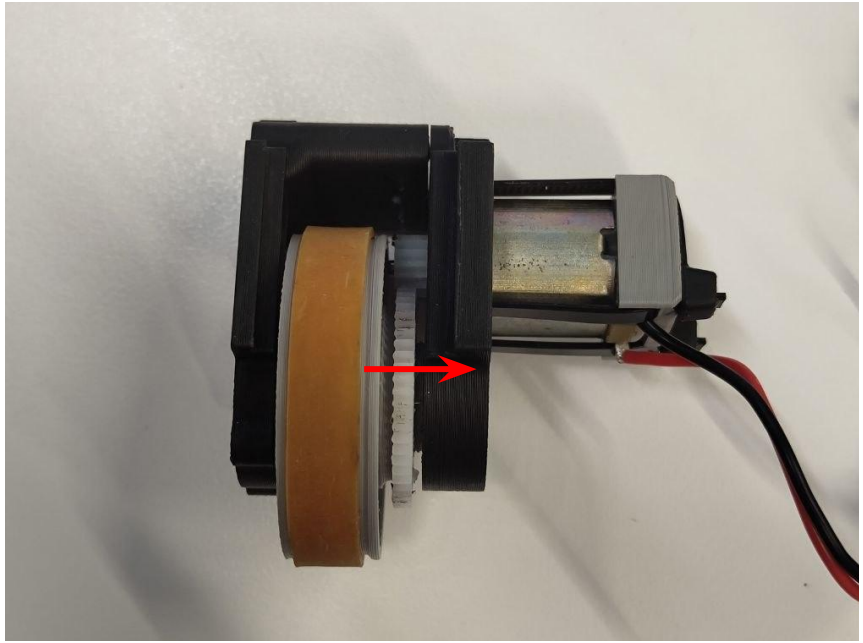

Press fit 17x30x7mm bearing on opposite side.

5.



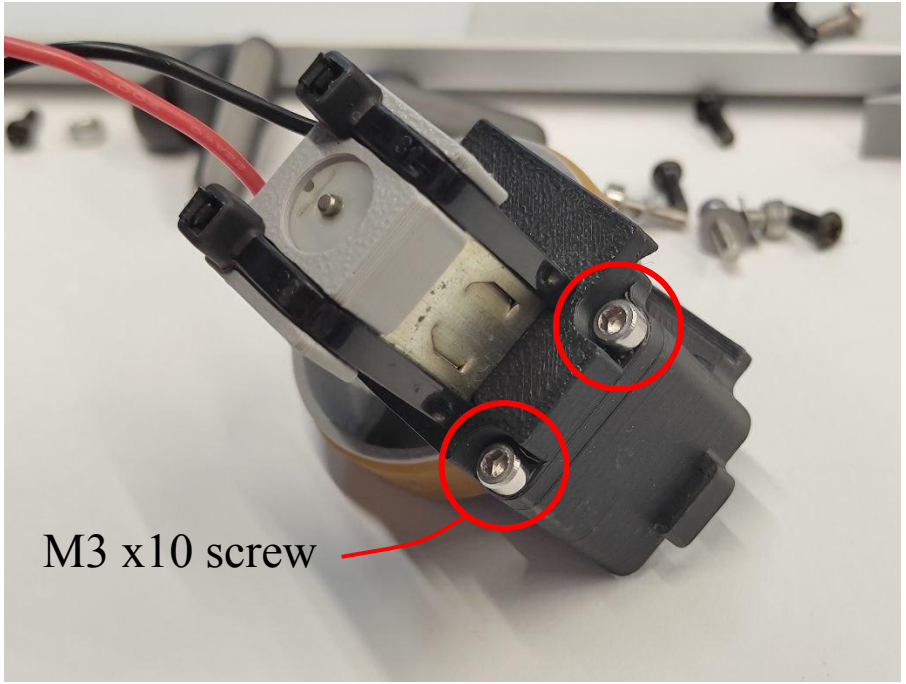
Press fit flywheel bearing side without gear into L-shaped mount.

6.



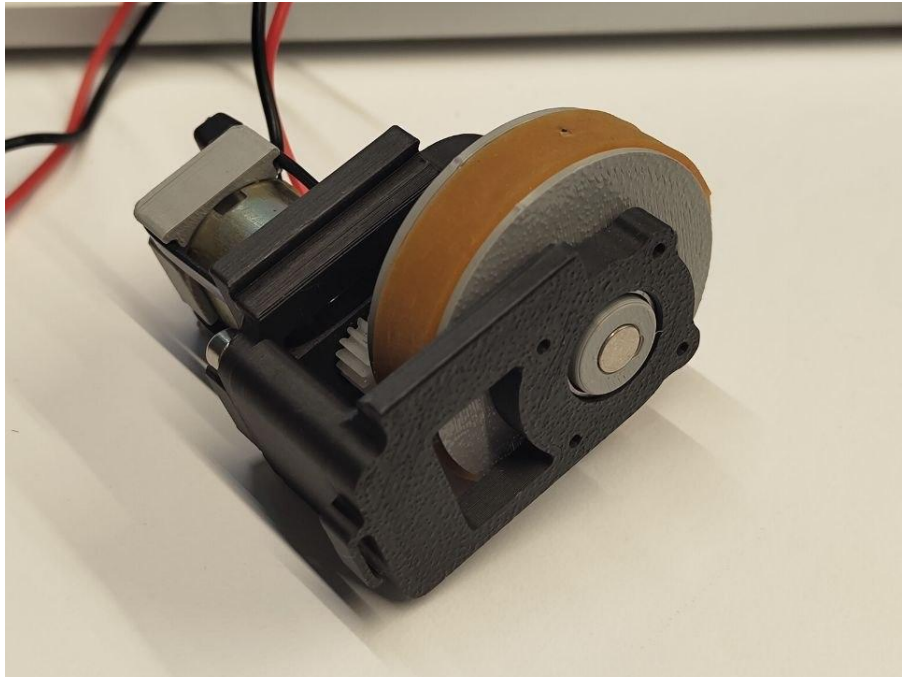
Press fit flywheel bearing on gear side into motor housing. The gear teeth should mesh.

7.



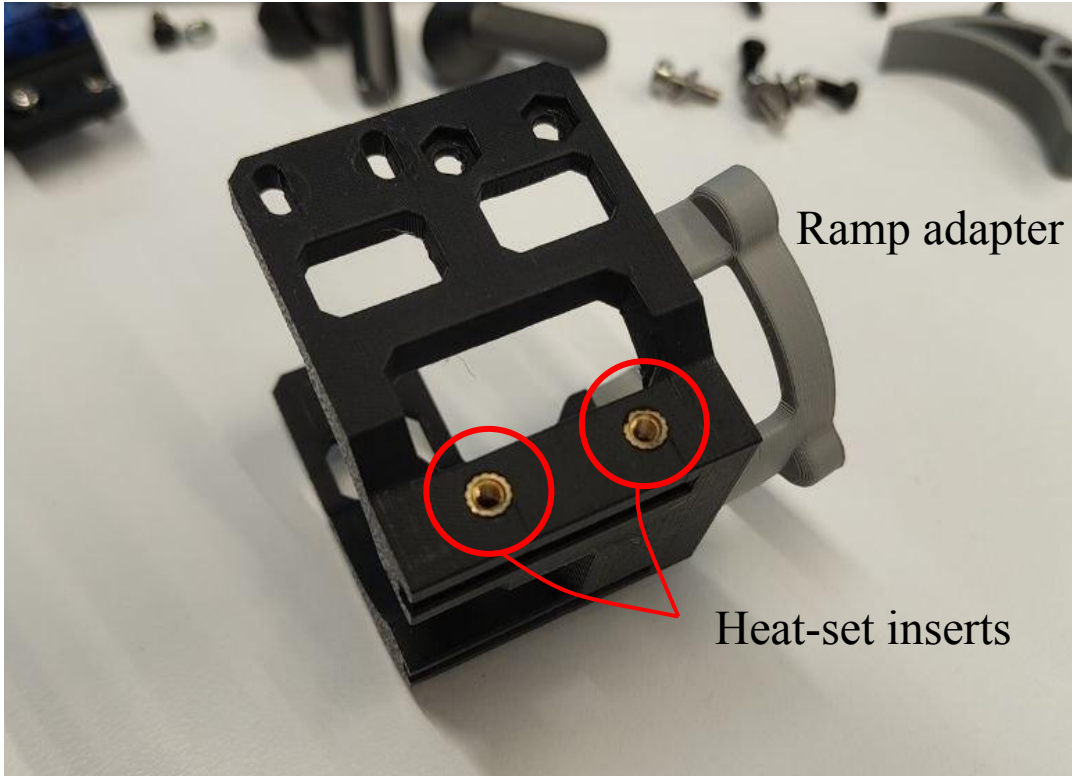

Secure L-shaped mount to motor housing using 2 M3 x10mm screws.

8.



Completed flywheel

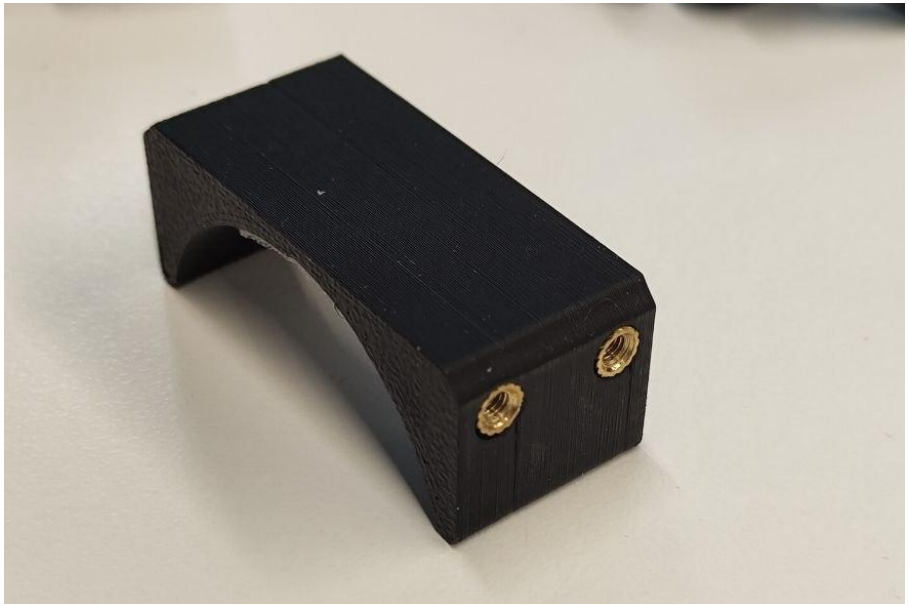
9.




Glue ramp adapter piece to ball hood.

Install 2 M3 x8mm heat-set inserts as shown.

**10.**

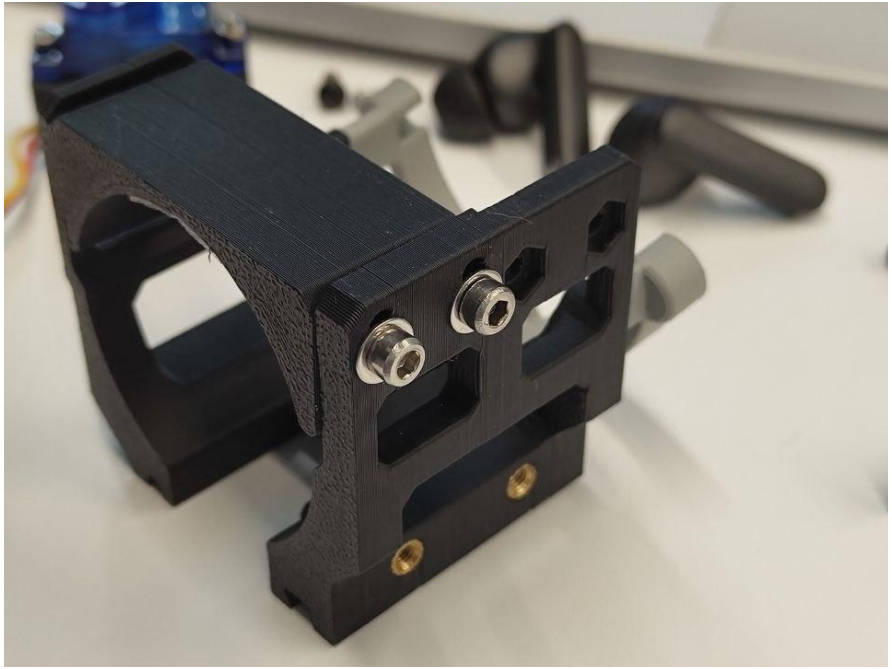


Install 2 M3 x8mm heat-set inserts in adjustable ball hood.

**11.**

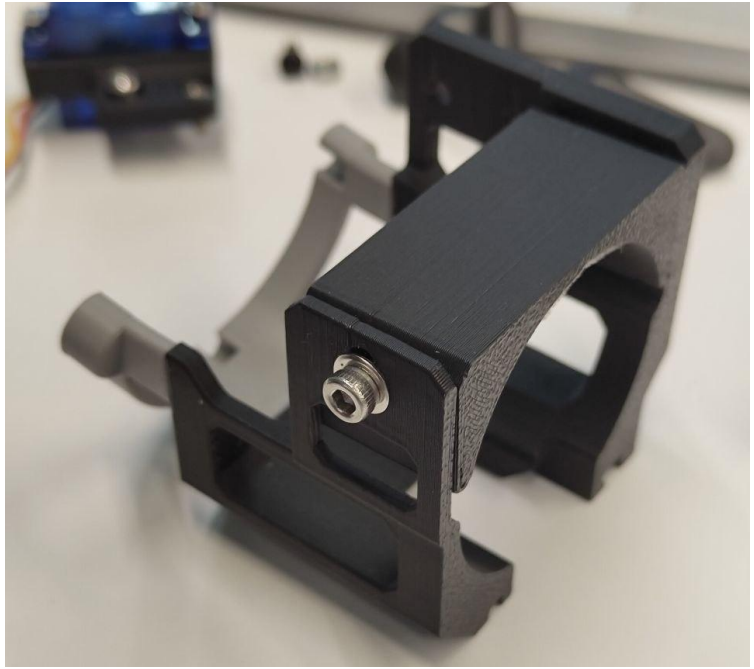
Install 1 M3 x8mm heat-set insert on other side.

**12.**



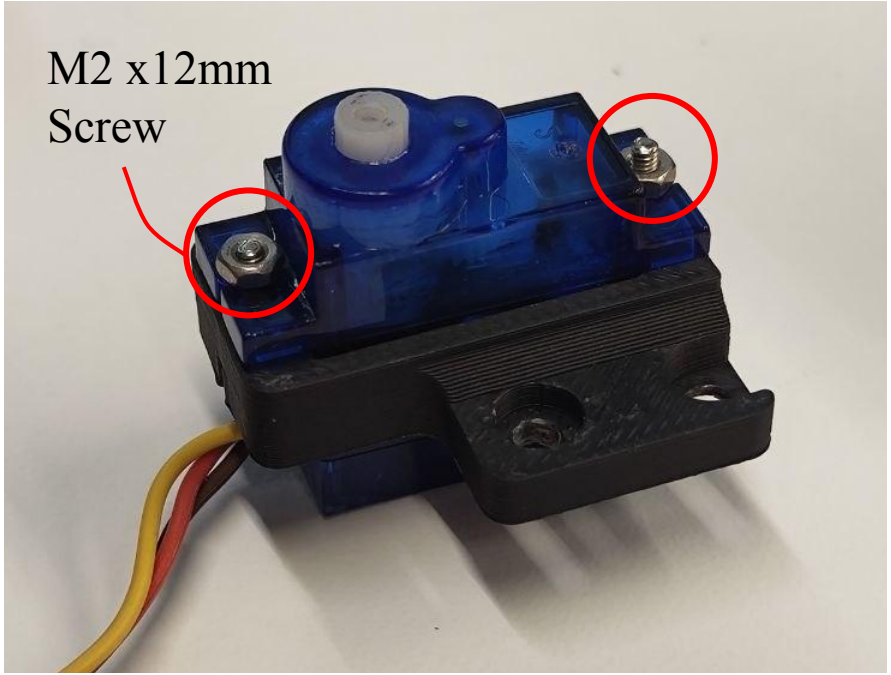
Attach adjustable hood using 2 M3 x10mm screws with washers. Loosen the screws to adjust hood height.

**13.**



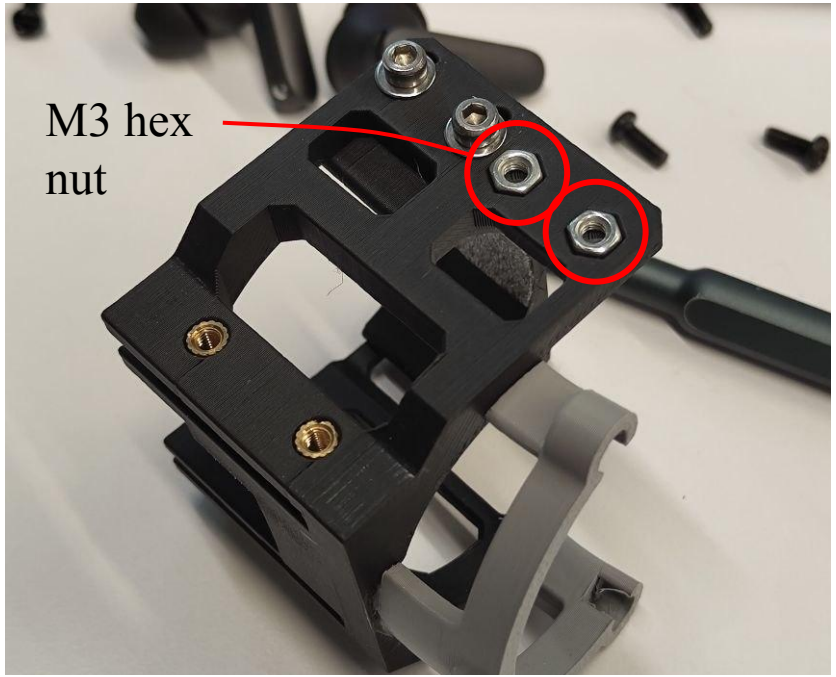
On the opposite side, attach with M3 x8mm screw with washer.

**14.**



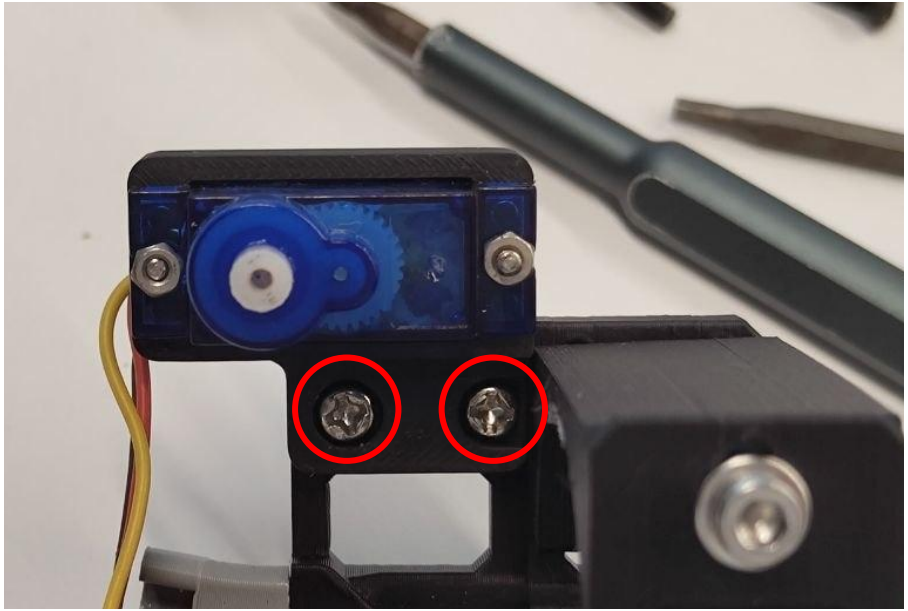

Slide SG90 servo into servo mount bracket and secure with 2 M2 x12mm screws.

**15.**



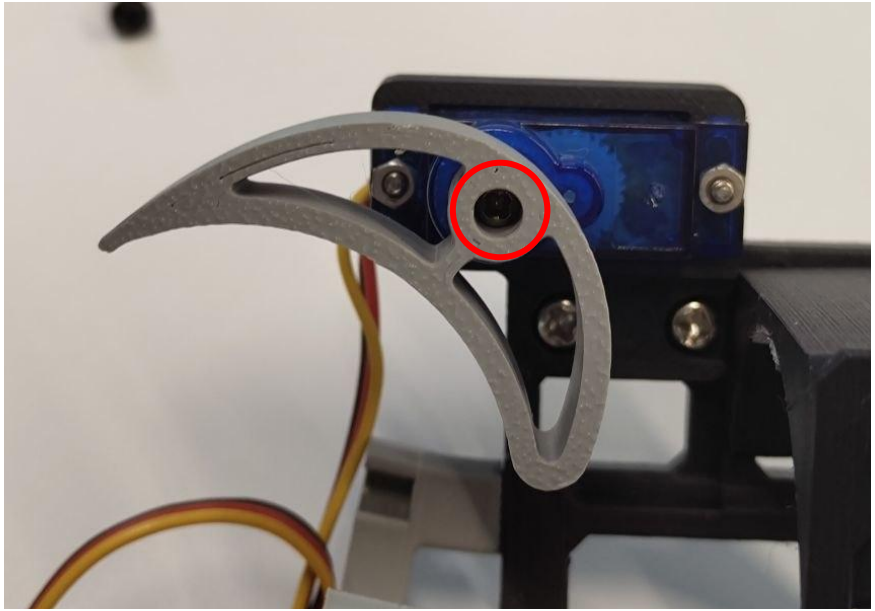

Press fit 2 M3 captured hex nuts to fasten the servo bracket.

**16.**



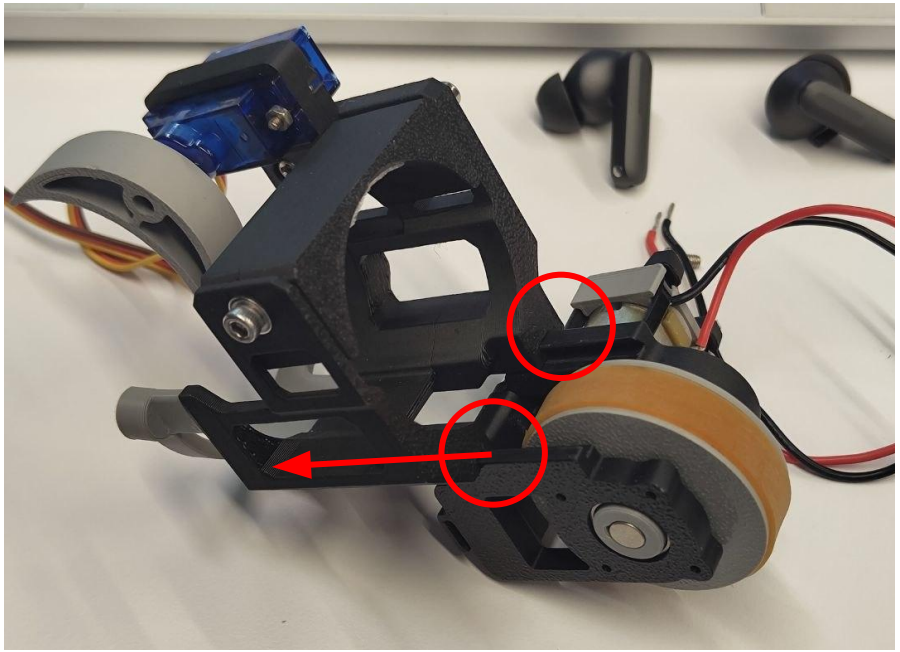
Fasten servo bracket to ball hood with 2 M3 x6mm screws.

**17.**



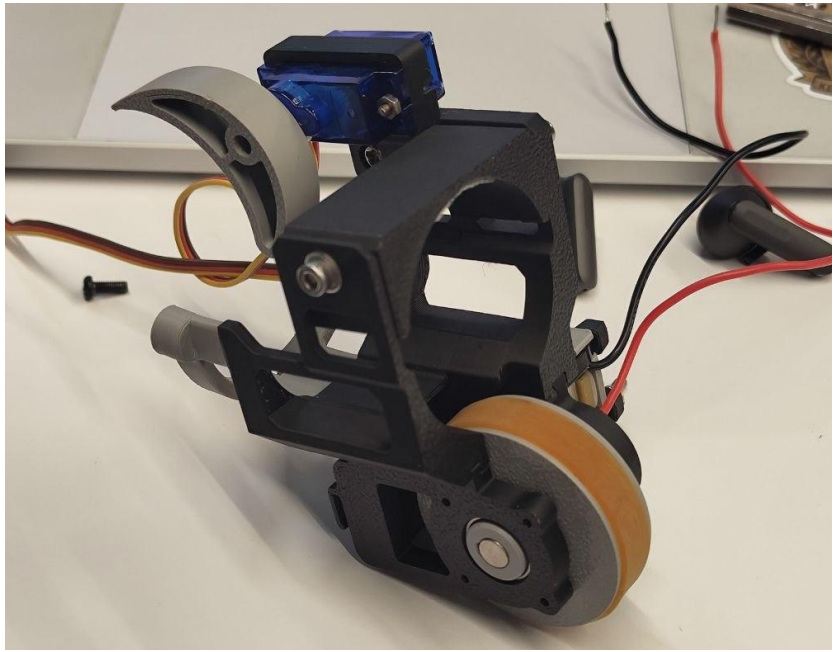
Attach ball indexer to SG90 servo and secure with M2 x8mm screw.

**18.**



Slide flywheel assembly into ball hood using dovetail joints.

**19.**

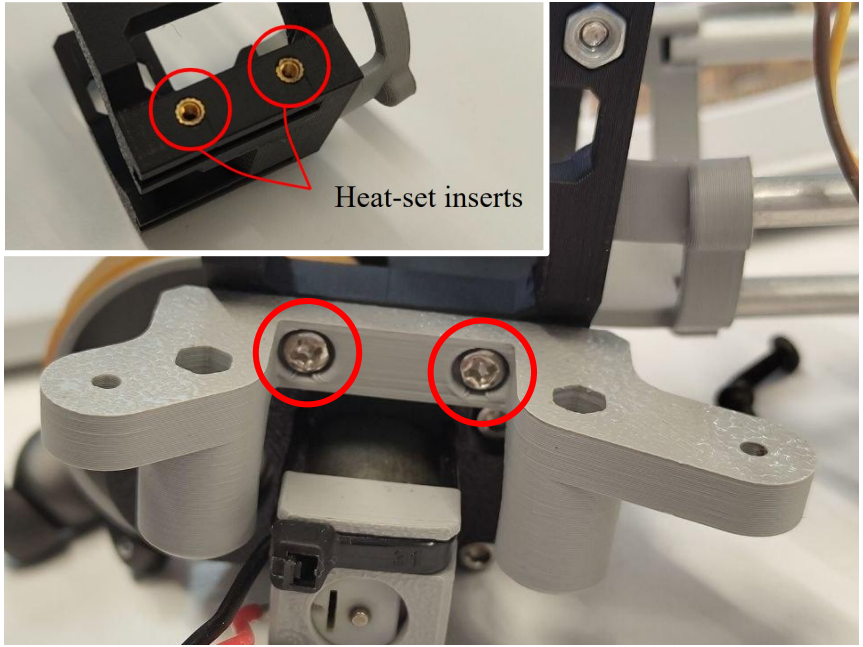


Fully seated assembly.

**20.**

Press fit 2 M3 captured hex nuts into flywheel assembly mounting piece.

**21.**

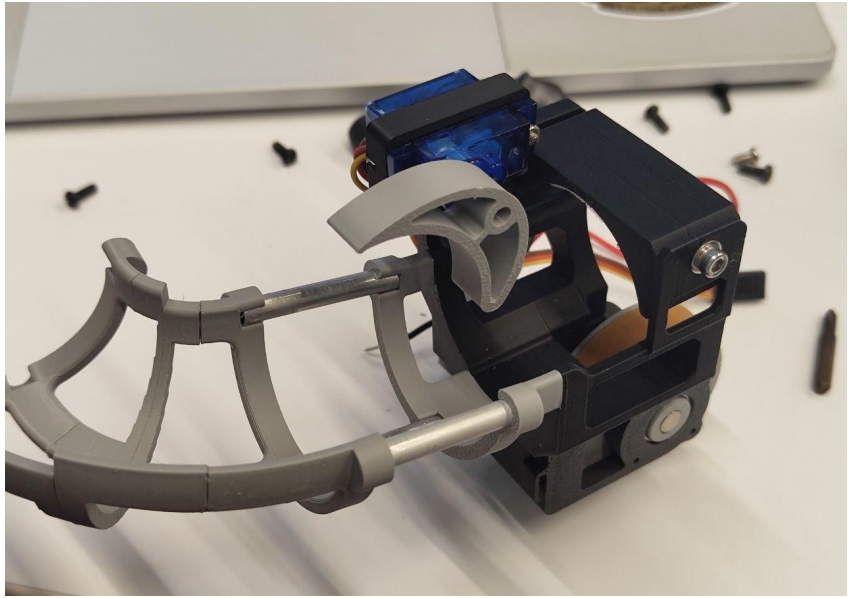



Screw mounting piece onto flywheel assembly, mating with heat-set inserts.

**22.**

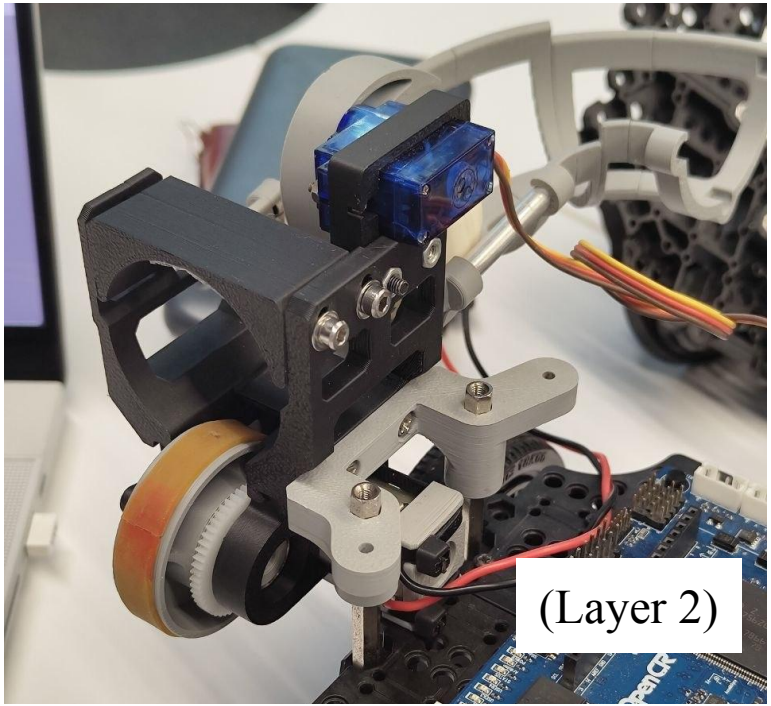
Glue 1 set of 90° ramp connectors and attach 3 6x35mm aluminium rods.

**23.**



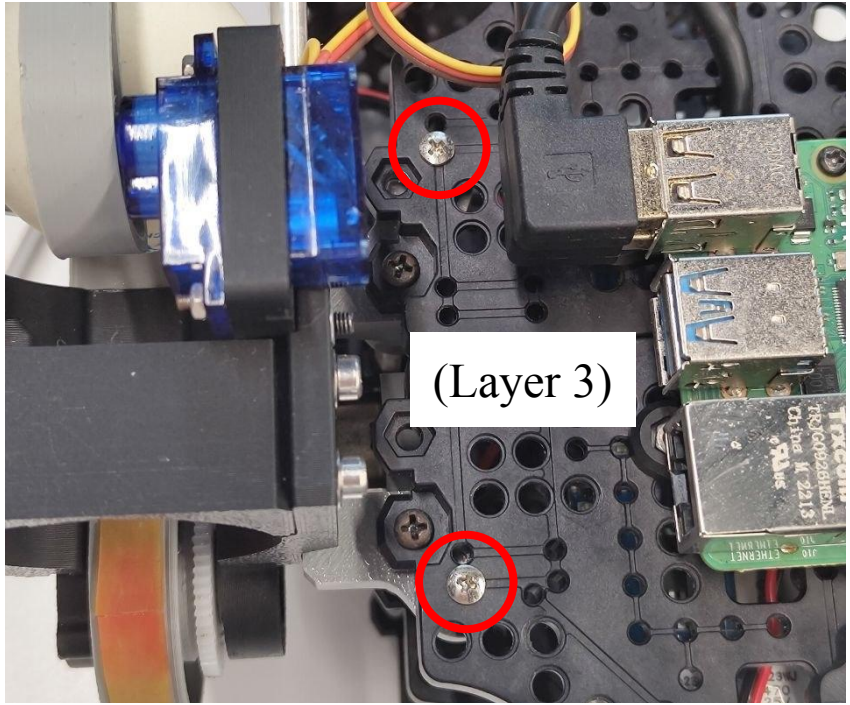
Attach short 90° ramp to ramp connector on ball hood.

24.



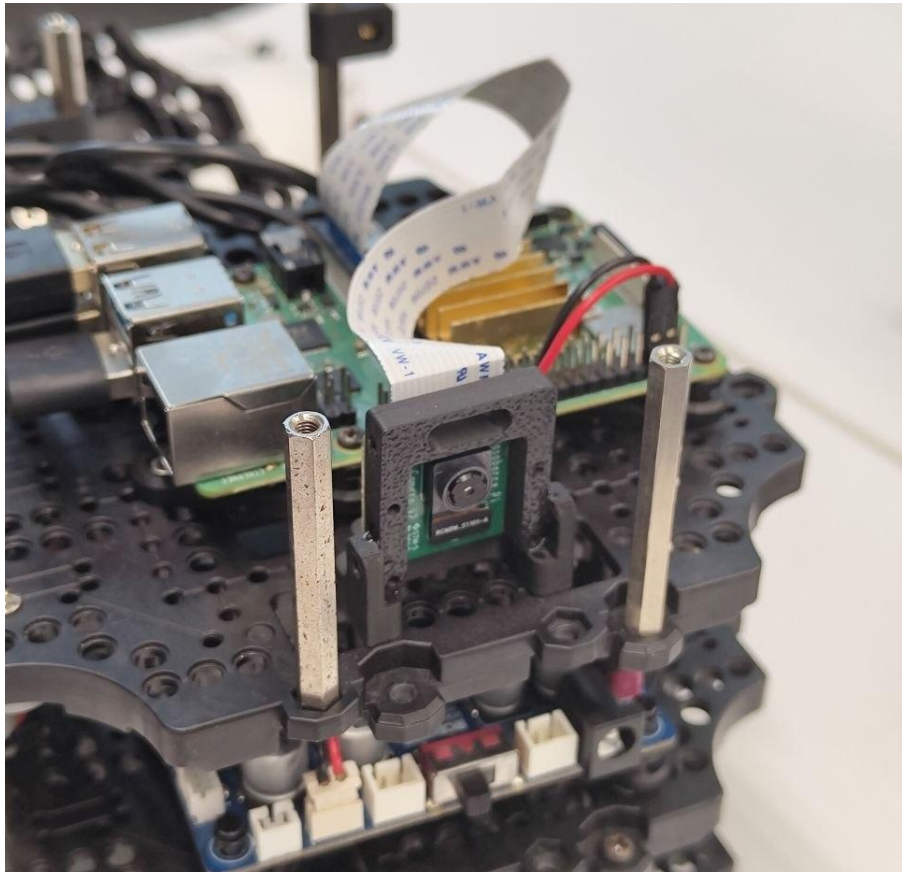

Slide flywheel mounting piece onto 45mm stand-offs on the second layer, to the right of the OpenCR.

25.




Attach the 3rd turtlebot layer, and use 2 M3 x16mm screws to secure.

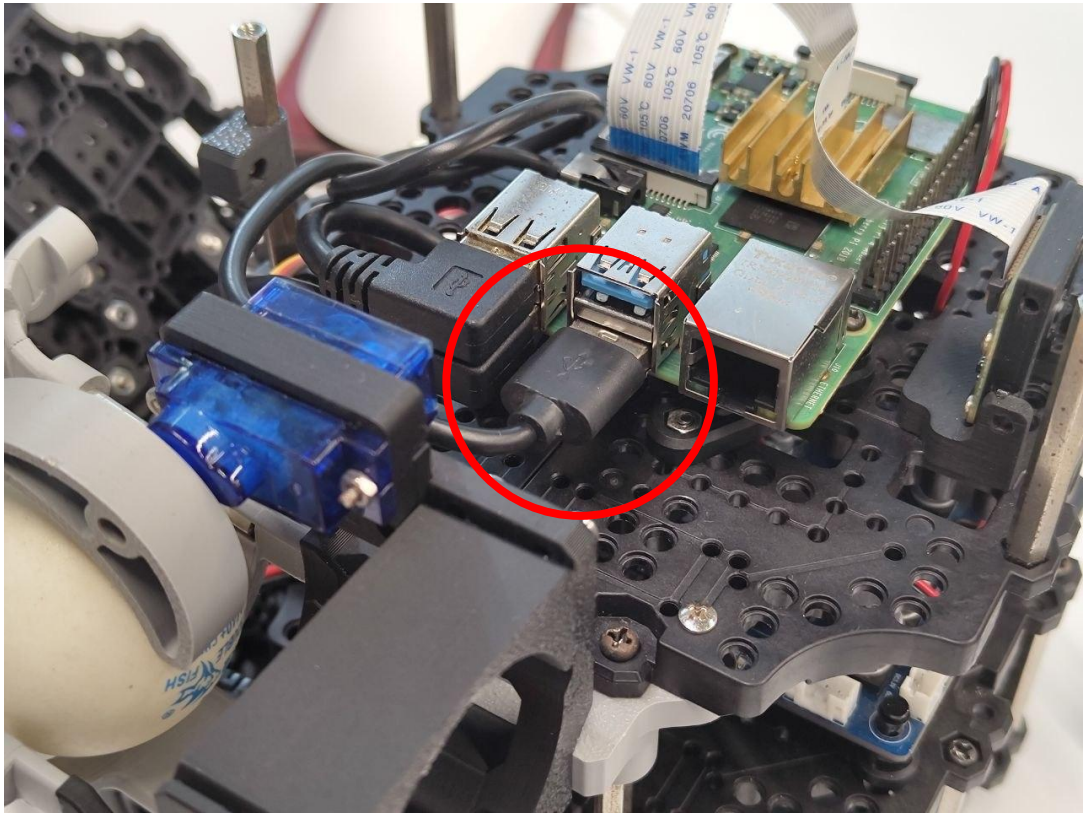
**26.**



Screw in the Raspberry Pi camera V2 mount using 2 M3 x12mm screws from the bottom of the plate.

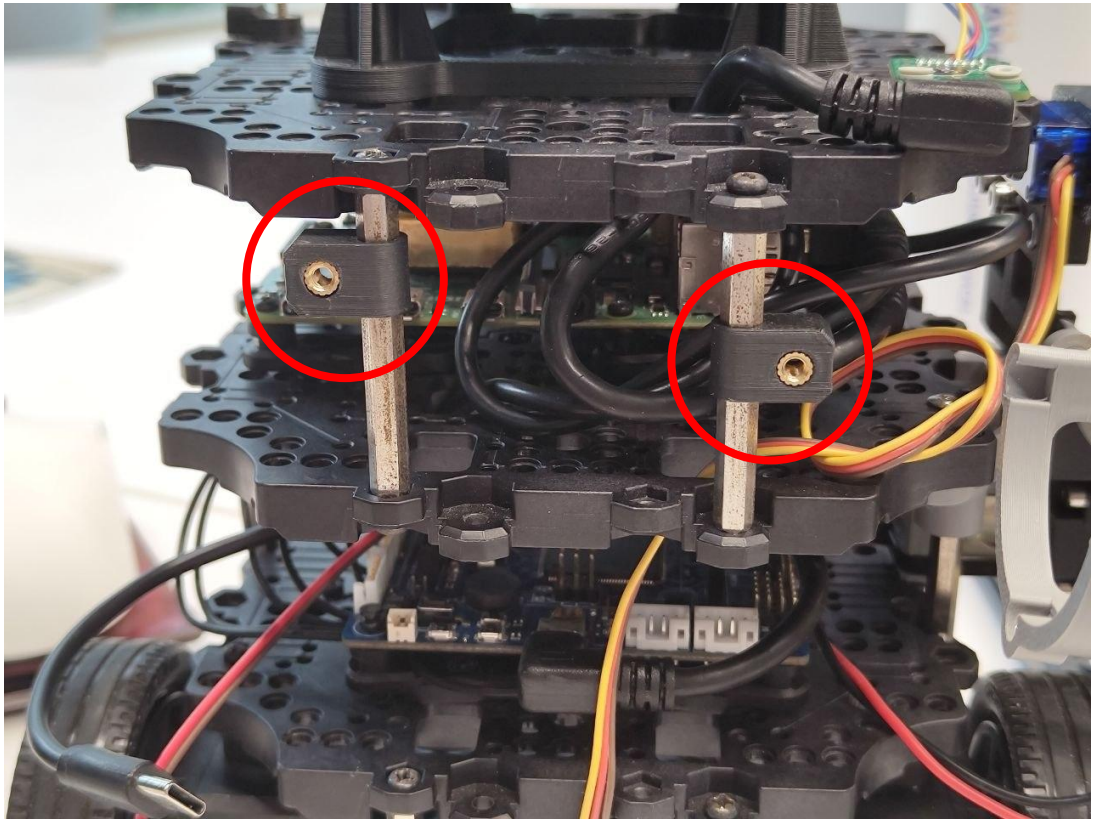
Connect the ribbon cable to the Raspberry Pi.

27.



Plug in USB-C cable for launcher controller module to Raspberry Pi.

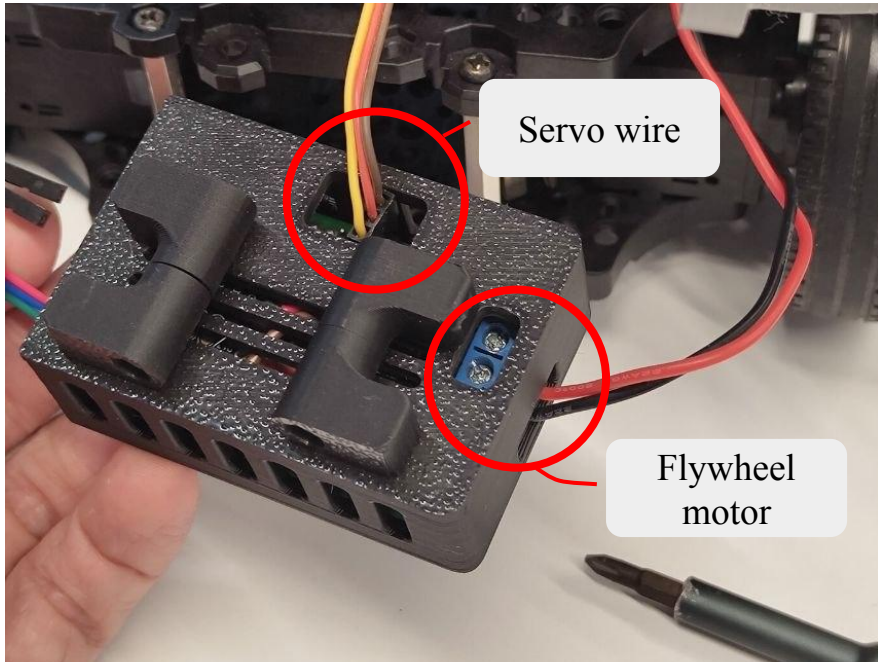
**28.**



Install heat-set insert for ramp mounting pieces.

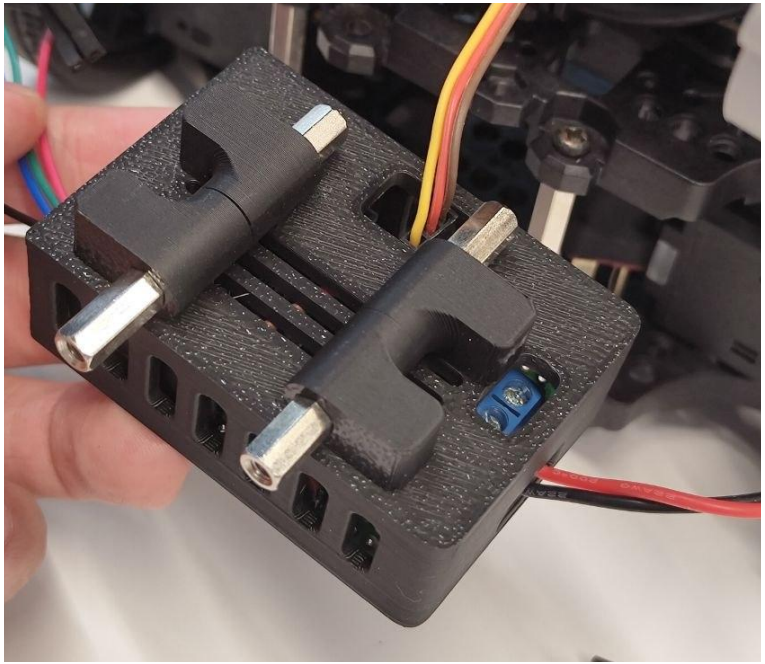
Slide in mounting points for ramp on hex stand-offs on the 3rd layer

29.



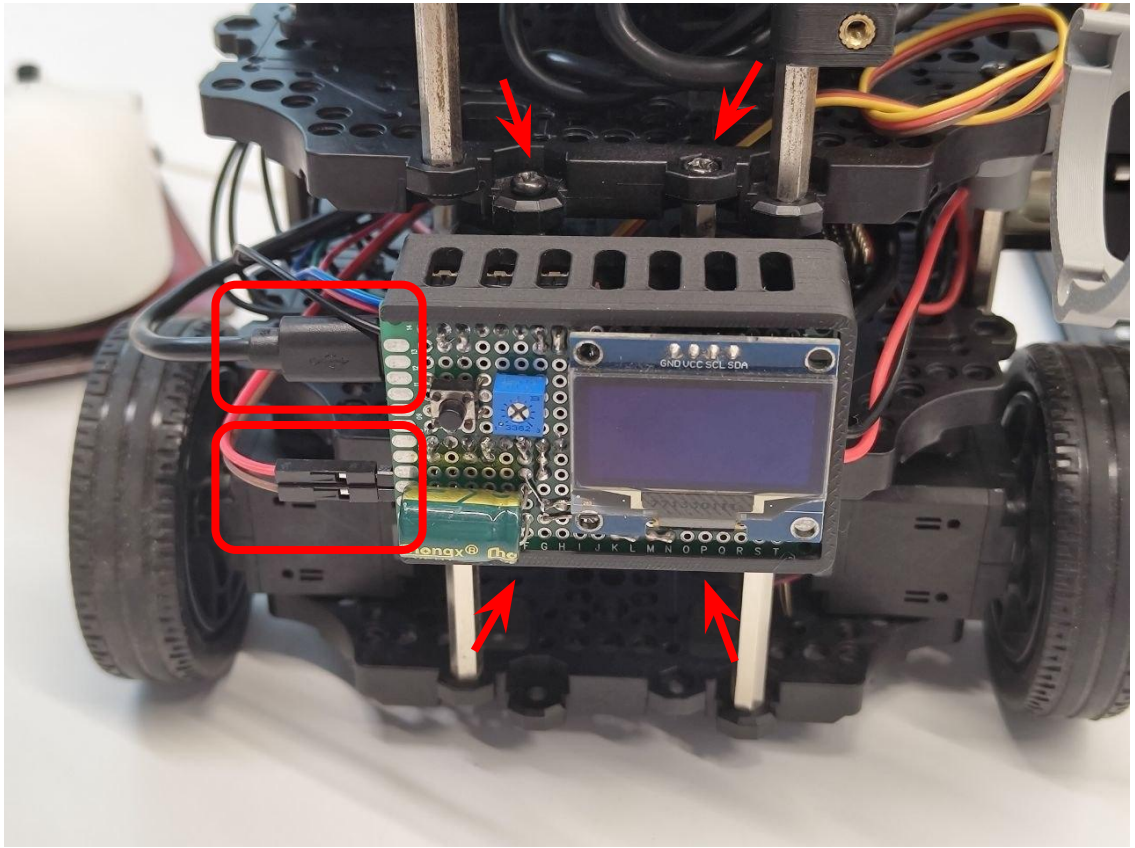

Screw in flywheel motor leads and plug servo wires into controller module.

30.



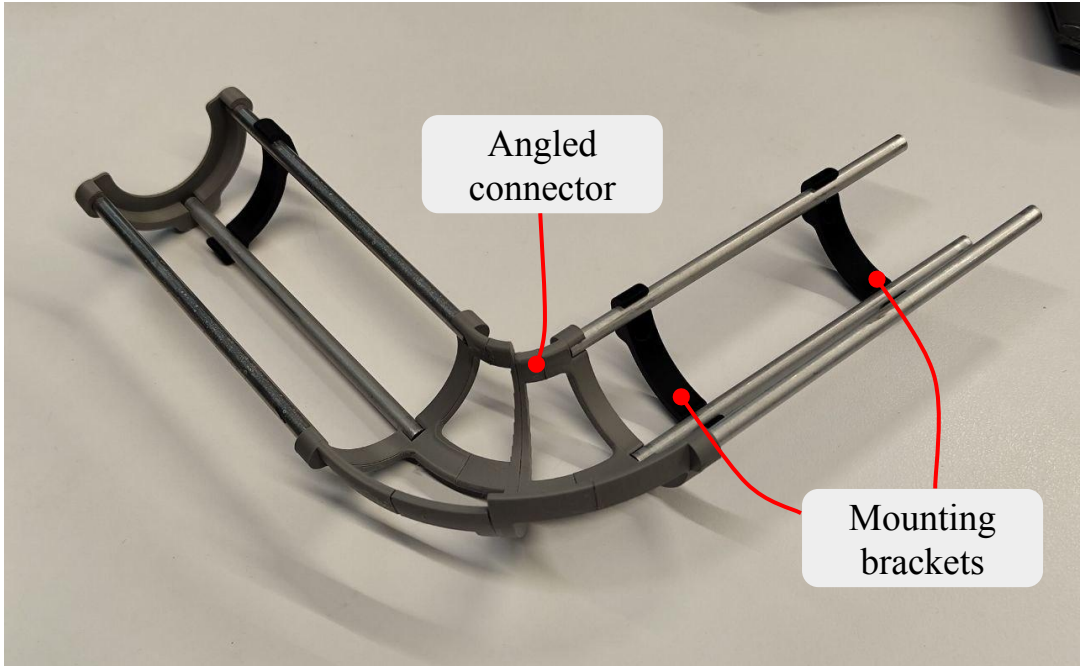
Insert 2 45mm hex stand-offs into controller module case.

31.




Install hex stand-offs between layers 2 and 3 with 4x Turtlebot M3 Black screws.

Plug USB-C cable into controller & voltage supply from buck converter.

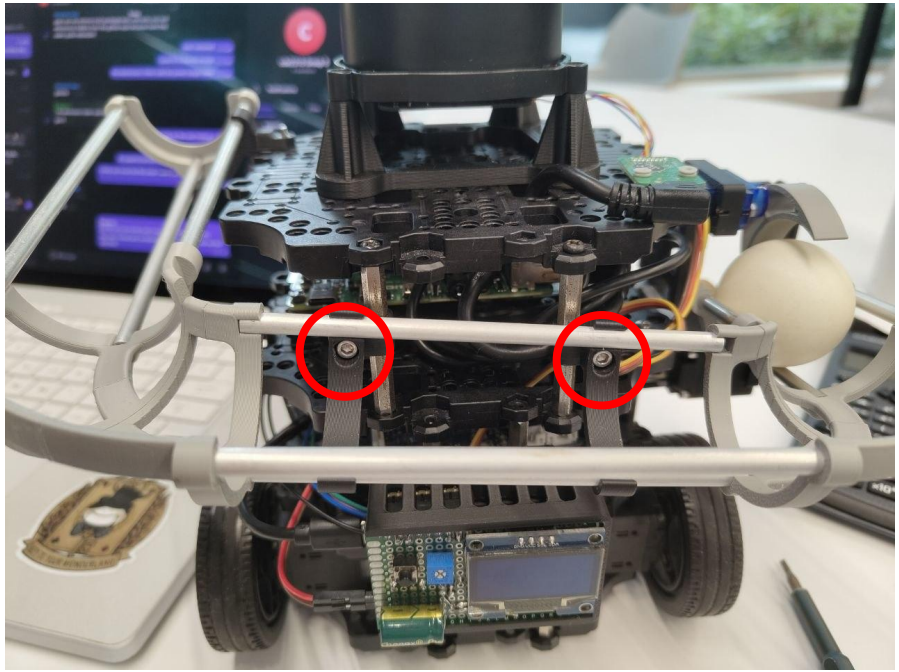



Fabricate rest of the ramp using 6x110mm aluminium rods and 1 set of angled connecetor glued together.

Attach the end piece to prevent balls from falling off.

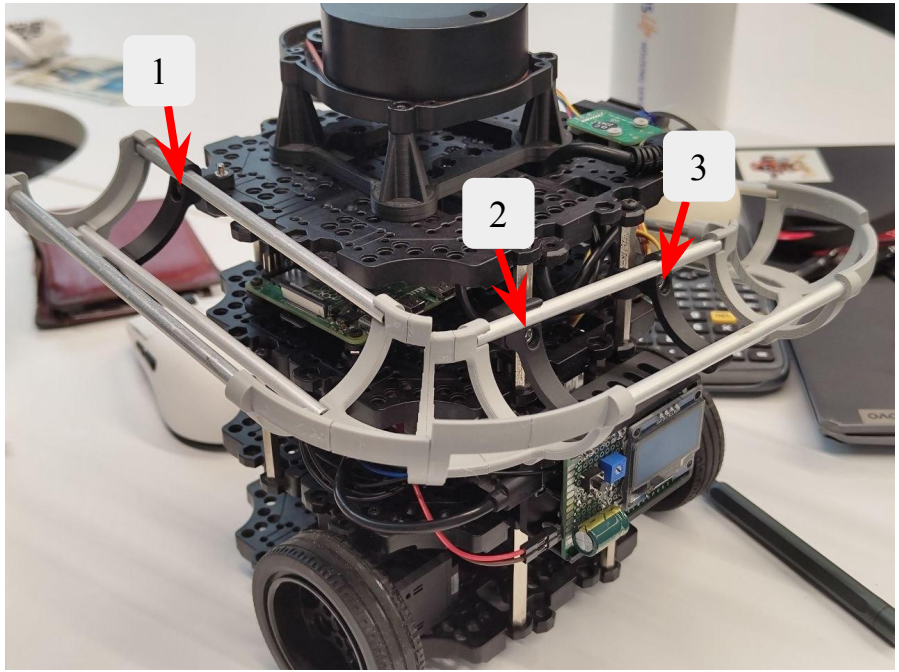
Clip on mounting brackets to attach to robot frame.

33.



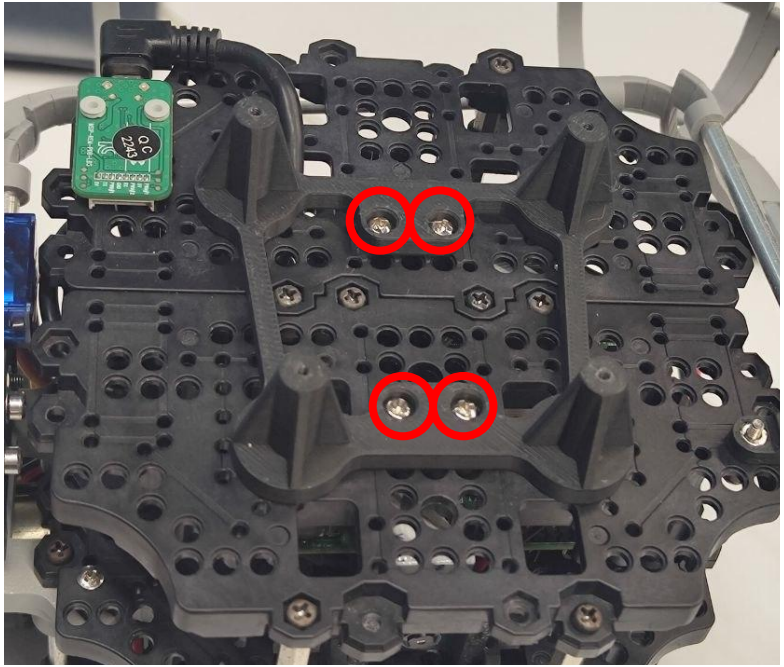
Attach ramp to the flywheel assembly rampc connector, then screw in mounting brackets to the frame connection points with M3 x6mm screws.

34.



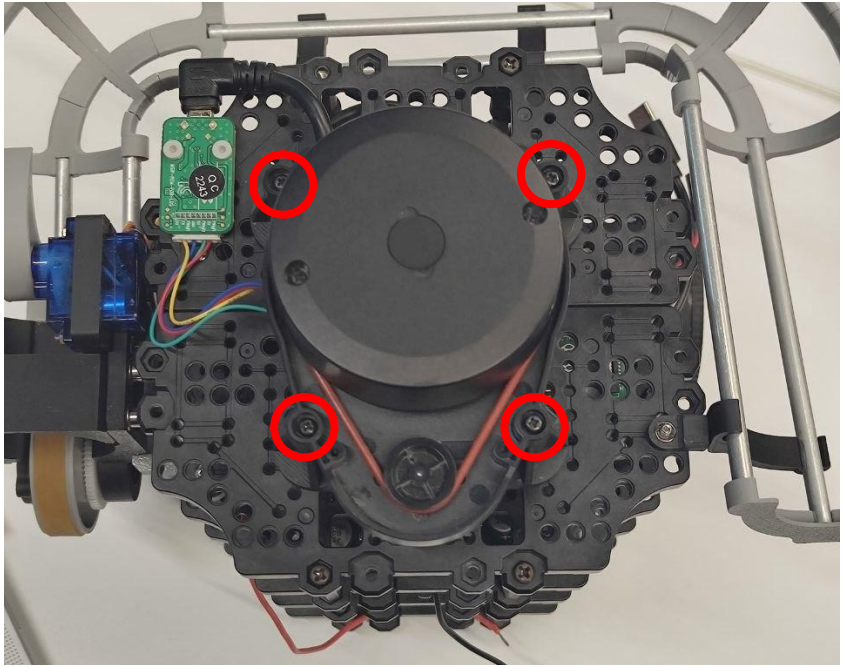

3 mounting points.

35.



Install LIDAR mount using 4 M3 x16mm screws, with a hex nut on the bottom of the plate.

36.



Mount LDS-02 using provided screws.